1 Comparison of MCOA and OMCOA

1.1 Simulation study

MCOA and OMCOA are compared on simulated data. $K = 3$ blocks $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ with respectively $p_1 = 10$, $p_2 = 15$, $p_3 = 20$ variables and $N = 100$ observations are generated from

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad k = 1, 2, 3,$$

where $\mathbf{U} \in \mathcal{M}_{N,R}(\mathbb{R})$ (column-centered) and $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ have orthonormal columns, $R = 3$, Σ_k is diagonal and $\mathbf{E}_k$ is a Gaussian noise term. We compare the estimation of $\Sigma_1, \ldots, \Sigma_K$ obtained with MCOA and OMCOA. The diagonal matrices depend on a parameter $t \in [0, 1]$:

$$\Sigma_k(t) = (1 - t)\,\Sigma_k^{(0)} + t\,\Sigma_k^{(1)},$$

where

- for $t = 0$, all the blocks share the same profile: $\Sigma_1^{(0)} = \Sigma_2^{(0)} = \Sigma_3^{(0)} = \mathrm{diag}(0.7, 0.5, 0.3)$;
- for $t = 1$, each block has its own dominant dimension of the common space $\mathbf{U}$ (permuted profiles): $\Sigma_1^{(1)} = \mathrm{diag}(0.7, 0.5, 0.3)$, $\Sigma_2^{(1)} = \mathrm{diag}(0.5, 0.3, 0.7)$, $\Sigma_3^{(1)} = \mathrm{diag}(0.3, 0.7, 0.5)$.

In both cases $\sum_r (\sigma_k^{(r)})^2 = 0.83$ for each block. Moreover, the singular values of the noise-free concatenated matrix are $\left(\sum_k (\sigma_k^{(r)}(t))^2\right)^{1/2}$: they are well separated for $t = 0$ and all equal to $\sqrt{0.83}$ for $t = 1$, since $\sum_k \Sigma_k^{(1)\,2} = 0.83\,\mathbf{I}_3$. The noise is scaled so that $\|\mathbf{E}_k\|^2 = \frac{0.10}{0.90}\,\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$, i.e. it accounts for about 10% of the variance of each block. We use $t \in \{0, 0.1, \ldots, 1\}$ and $B = 1000$ replications.

At each replication, the estimated axes are reordered by the permutation

$$\hat{\pi} = \arg\max_{\pi \in \mathfrak{S}_R} \sum_{r=1}^{R} \left|\cos\left(\mathbf{u}^{(r)}, \hat{\mathbf{u}}^{(\pi(r))}\right)\right|,$$

the absolute value accounting for the sign indeterminacy of the axes. The accuracy of the estimation of $\Sigma_1, \ldots, \Sigma_K$ is assessed by the mean absolute error over the K blocks and the R dimensions:

$$\mathrm{MAE} = \frac{1}{KR} \sum_{k=1}^{K} \sum_{r=1}^{R} |\hat{\sigma}_k^{(\hat{\pi}(r))} - \sigma_k^{(r)}|.$$

1.2 Results

Figures 1 and 2 show the estimates obtained with MCOA and OMCOA for $t = 0$ and $t = 1$ respectively. As expected, both methods give very close estimates when the blocks share a common profile (Figure 1). When the profiles are permuted, OMCOA outperforms MCOA in terms of both bias and precision (Figure 2). In both scenarios, the OMCOA estimates show very low variability.

Figure 3 shows the empirical distribution of the percentage of variance of the concatenated matrix $\mathbf{X}$ explained by the R dimensions,

$$V = 100 \times \frac{\sum_{k,r} \left(\hat{\sigma}_k^{(r)}\right)^2}{\|\mathbf{X}\|^2}, \qquad \|\mathbf{X}\|^2 = \sum_{k=1}^{K} \|\mathbf{X}_k\|^2,$$

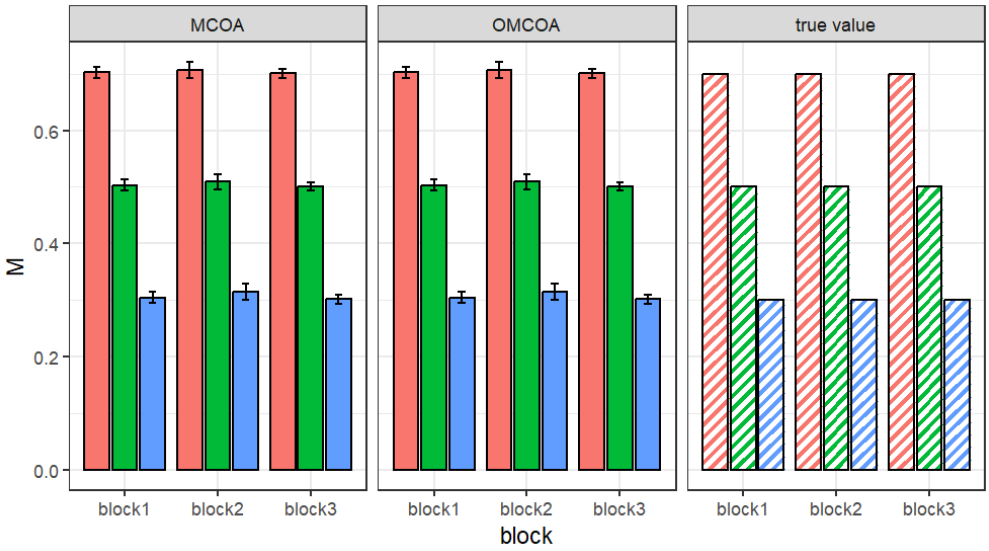


Figure 1: Common profile ($t = 0$): median and 10–90% empirical quantiles of $\hat{\sigma}_k^{(\hat{\pi}(r))}$ over the $B = 1000$ replications (red: dim 1, green: dim 2, blue: dim 3).

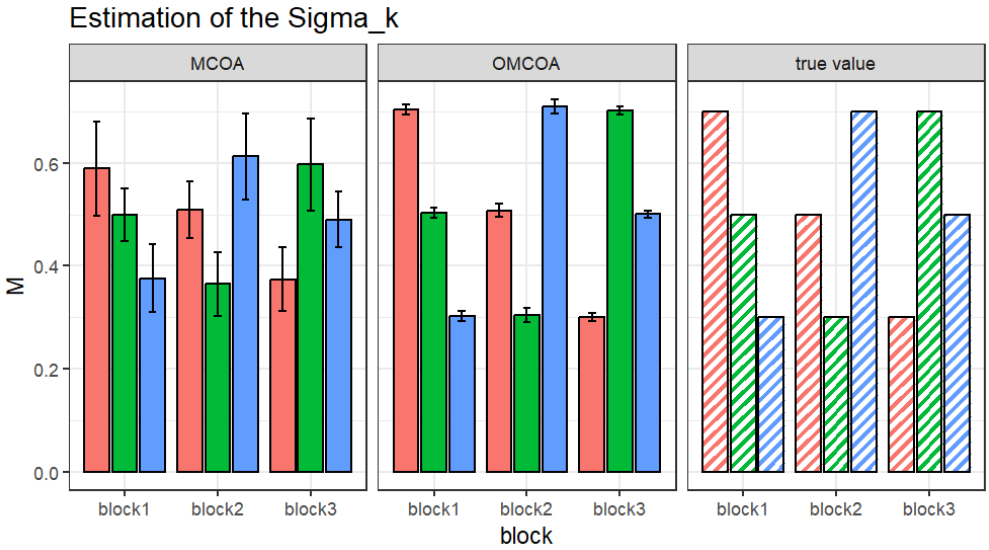


Figure 2: Permuted profiles ($t = 1$): median and 10–90% empirical quantiles of $\hat{\sigma}_k^{(\hat{\pi}(r))}$ over the $B = 1000$ replications (red: dim 1, green: dim 2, blue: dim 3).

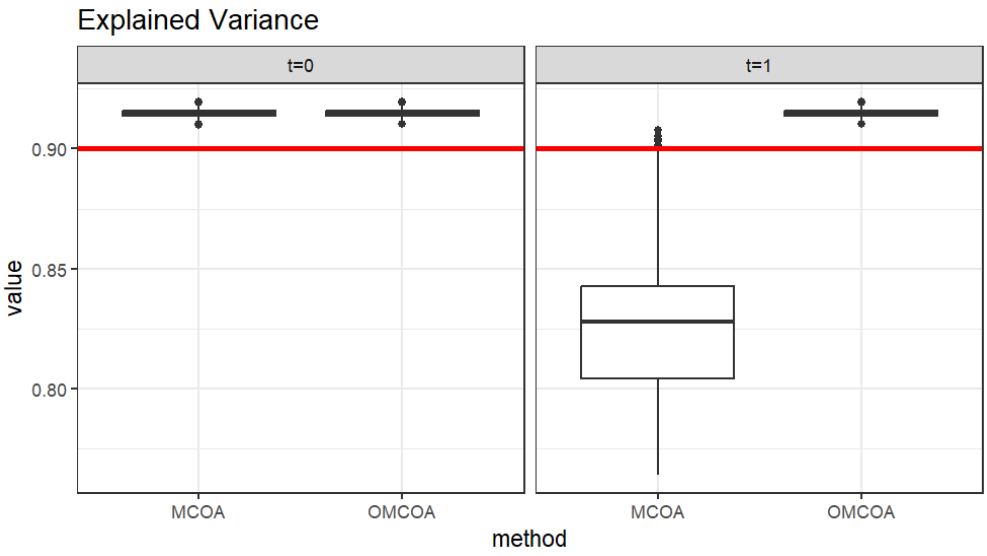


Figure 3: Distribution of the percentage of explained variance V over the $B = 1000$ replications.

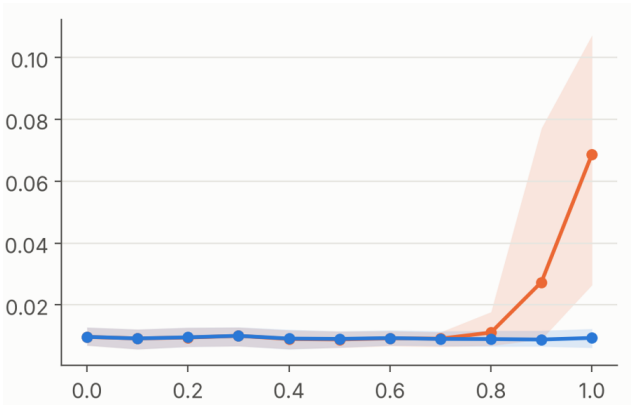


Figure 4: Median and 10–90% empirical quantiles of the MAE as a function of t ($B = 1000$ replications).

to be compared with the percentage of variance due to the signal, which is about 90% by construction. OMCOA slightly exceeds this value in all cases, since part of the noise is captured by the fitted dimensions, whereas MCOA clearly underestimates it when the profiles are permuted ($t = 1$).

Figure 4 shows the distribution of the MAE as a function of t. MCOA and OMCOA coincide as long as the singular values of $\mathbf{X}$ are well separated; MCOA breaks down when they get close ($t \geq 0.8$). Indeed, MCOA starts with the SVD of $\mathbf{X}$: when the singular values are close, $\mathbf{u}^{(1)}$ is an arbitrary direction of span($\mathbf{U}$) driven by the noise, and this error propagates to the next dimensions through the deflation. OMCOA fits all the blocks simultaneously, so that the block-specific profiles identify $\mathbf{U}$. Figure 5 illustrates the resulting misestimation of the axes $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(R)}$ by MCOA for $t \geq 0.8$.

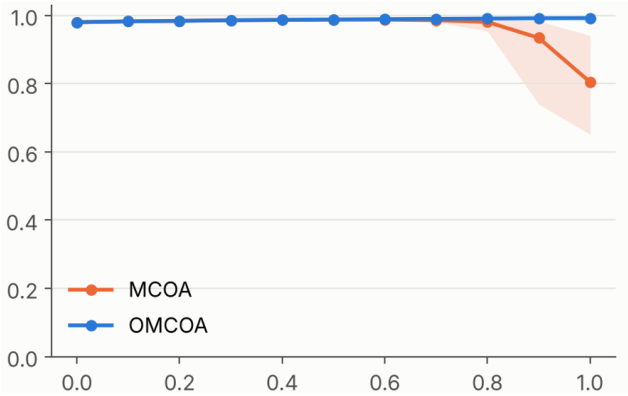


Figure 5: Median and 10–90% empirical quantiles of the worst axis recovery $\min_r |\cos(\mathbf{u}^{(r)}, \hat{\mathbf{u}}^{(\hat{\pi}(r))})|$ as a function of t ($B = 1000$ replications).